

cases), whose LOS was therefore substantially shorter than it would be if they hadn't died.

- Eisenstein et al. (2001) estimated 10-year costs of \$44,663, in 1997\$ (using a three percent discount rate), or \$85,052 in 2015\$ for MI patients, using statistical prediction (regression) models to estimate inpatient costs. Only inpatient costs (physician fees and hospital costs) were included. Estimates from Eisenstein et al. are included in Appendix I.4 as they are not used in EPA impact analyses.
- Russell et al. (1998) estimated first-year direct medical costs of treating nonfatal MI of \$15,540 (in 1995\$), and \$1,051 annually thereafter. Converting to year 2015\$, that would be \$38,253 for a 5-year period, using a three percent discount rate, or \$36,167, using a seven percent discount rate.

The age group-specific estimates of opportunity cost over a five-year period are combined with the medical cost estimates from each of the three studies listed above. Because opportunity costs are derived for each of five age groups, there are $3 \times 5 = 15$ unit values for each of 2 discount rates, or 30 unit values available for use in BenMAP. These are given in Table I-6 below.

Note that we were unable to achieve complete consistency, unfortunately, because of limitations in the input studies. For example, although we calculated opportunity costs over a five-year period using a 3 percent and a 7 percent discount rate, we were not able to do the same for medical costs, except for the medical costs estimated by Russell et al. (in which they estimate an annual cost). Wittels et al. appear to have used no discounting in their estimate; Eisenstein et al. used a 3 percent discount rate. Similarly, although almost all cost estimates (opportunity costs and medical costs) are for a 5-year period, the medical cost estimate reported by Eisenstein et al. is for a 10-year period. There was no reasonable method for inferring from that study what costs over a 5-year period would be.

Table I-6. Additional Unit Values Available for Myocardial Infarction

Basis of Estimate	Age Range		Medical Cost *	Opportunity Cost **	Total Cost
	Min	Max			
COI: 5 yrs med, 5 yrs wages, 3% DR, Wittels (1990)	0	24	\$187,530	\$0	\$187,530
	25	44	\$187,530	\$13,301	\$200,831
	45	54	\$187,530	\$19,604	\$207,134
	55	65	\$187,530	\$113,316	\$300,846
	66	99	\$187,530	\$0	\$187,530
COI: 5 yrs med, 5 yrs wages, 3% DR, Russell (1998)	0	24	\$38,253	\$0	\$38,253
	25	44	\$38,253	\$13,301	\$51,554
	45	54	\$38,253	\$19,604	\$57,857
	55	65	\$38,253	\$113,316	\$151,569
	66	99	\$38,253	\$0	\$38,253
COI: 10 yrs med, 5 yrs wages,	0	24	\$85,052	\$0	\$85,052

Basis of Estimate	Age Range		Medical Cost *	Opportunity Cost **	Total Cost
	Min	Max			
3% DR, Eisenstein (2001)	25	44	\$85,052	\$13,301	\$98,353
	45	54	\$85,052	\$19,604	\$104,656
	55	65	\$85,052	\$113,316	\$198,368
	66	99	\$85,052	\$0	\$85,052
COI: 5 yrs med, 5 yrs wages, 7% DR, Wittels (1990)	0	24	\$187,530	\$0	\$187,530
	25	44	\$187,530	\$11,908	\$199,438
	45	54	\$187,530	\$17,552	\$205,082
	55	65	\$187,530	\$101,451	\$288,981
	66	99	\$187,530	\$0	\$187,530
COI: 5 yrs med, 5 yrs wages, 7% DR, Russell (1998)	0	24	\$36,167	\$0	\$36,167
	25	44	\$36,167	\$11,908	\$48,075
	45	54	\$36,167	\$17,552	\$53,719
	55	65	\$36,167	\$101,451	\$137,618
	66	99	\$36,167	\$0	\$36,167
COI: 10 yrs med, 5 yrs wages, 7% DR, Eisenstein (2001)	0	24	\$85,052	\$0	\$85,052
	25	44	\$85,052	\$11,908	\$96,960
	45	54	\$85,052	\$17,552	\$102,604
	55	65	\$85,052	\$101,451	\$186,503
	66	99	\$85,052	\$0	\$85,052

* An average of the 5-year costs estimated by Wittels et al. (1990) and Russell et al. (1998). Note that Wittels et al. appears not to have used discounting in deriving a 5-year cost of \$187,530; Russell et al. estimated first- year direct medical costs and annual costs thereafter. The resulting 5-year cost is \$38,253, using a 3% discount rate, and \$36,167, using a 7% discount rate. Medical costs were inflated to 2015\$ using CPI for medical care.

** From Cropper and Krupnick (1999). Present discounted value of 5 yrs of lost earnings, at 3% and 7% discount rate, adjusted from 1977\$ to 2015\$ using CPI-U "all items".

I.4.2 Acute Bronchitis in Children

Estimating WTP to avoid a case of acute bronchitis is difficult for several reasons. First, WTP to avoid acute bronchitis itself has not been estimated. Estimation of WTP to avoid this health endpoint therefore may be based on estimates of WTP to avoid symptoms that occur with this illness. Second, a case of acute bronchitis may last more than one day, whereas it is a day of avoided symptoms that is typically valued. Finally, the C-R function used in the benefit analysis for acute bronchitis was estimated for children, whereas WTP estimates for those symptoms associated with acute bronchitis were obtained from adults.

Three unit values are available in BenMAP for acute bronchitis in children. In previous benefit analyses, EPA used a unit value of \$81.63 (adjusted to 2015\$). This is the midpoint between a low estimate and a high estimate. The low estimate is the sum of the midrange values recommended by IEc (1994) for two symptoms believed to be associated with acute bronchitis: coughing and chest tightness. The high estimate was taken to be twice the value of a minor respiratory restricted activity day. For a more complete description of the derivation of this estimate, see Abt Associates (2000, p. 4-

30). The above unit value assumes that an episode of acute bronchitis lasts only one day. However, this is generally not the case, as acute bronchitis typically lasts 6 or 7 days. To generate a unit value representative of an entire acute bronchitis episode, the original unit value of \$81.63 was multiplied by 6 or 7. A unit value of \$490 ($=\81.63×6) was therefore derived.

As discussed the epidemiological study relating air pollution to the incidence of acute bronchitis referred to children specifically. The value of an avoided case should therefore be WTP to avoid a case in a child, which may be different from WTP to avoid a case in an adult. Recent work by Dickie and Ulery (2002) suggests, in fact, that parents are generally willing to pay about twice as much to avoid sickness in their children as in themselves. In one of several models they estimated, the natural logarithm of parents' WTP was related both to the number of symptom-days avoided and to whether it was their child or themselves at issue. Dickie and Ulery noted that "experiencing all of the symptoms [considered in their study - cough and phlegm, shortness of breath/wheezing, chest pain, and fever] for 7 days, or 28 symptom-days altogether, is roughly equivalent to a case of acute bronchitis ..." Using this model, and assuming that a case of acute bronchitis can be reasonably modeled as consisting of 28 symptom-days, we estimated parents' WTP to avoid a case of acute bronchitis in a child to be \$529. This is the third unit value available in BenMAP.

The mean household income among participants in the Dickie and Ulery CV survey was slightly higher than the national average. We therefore adjusted all WTP estimates that resulted from their models downward slightly, using an income elasticity of WTP of 0.147, the average of the income elasticities estimated in the four models in the study. The adjustment factor thus derived was 0.9738. Estimates for Acute Bronchitis are available in Table I-5.

I.4.3 Minor Restricted Activity Days (MRADs)

See Appendix H, Section H.3.3 for more information.

In addition to the estimate of WTP to avoid a MRRAD used in EPA benefits analyses, a second unit value is based on Model 1, Table III in Dickie and Ulery (2002). This model estimates the natural logarithm of parents' WTP to avoid symptoms as a linear function of the natural logarithm of the number of symptom-days avoided and whether or not the person avoiding the symptoms is the parent or the child. The unit value derived from this model, assuming that an MRAD consists of one day of 3 symptoms in an adult, is \$138.

The estimate for the MRADs that is not used in EPA benefits analyses can be found in Table I-5.

I.4.4 Asthma Exacerbation

Table I-7 below describes the unit values for Asthma-related Acute Symptoms and Illnesses included in BenMAP but not used by the EPA for their regulatory impact analyses. All unit values for this section are summarized in Table I-7.

Several respiratory symptoms in asthmatics or characterizations of an asthma episode have been associated with exposure to air pollutants. All of these can generally be taken as indications of an asthma exacerbation (“asthma attack”) when they occur in an asthmatic. BenMAP therefore uses the same set of unit values for all of the variations of “asthma exacerbation” that appear in the epidemiological literature.

In Rowe and Chestnut (1986), the mean of the four average WTPs is \$32 (1990 \$), or \$59 in 2015\$. The uncertainty surrounding this estimate was characterized by a continuous uniform distribution on the range defined by the lowest and highest of the four average WTP estimates from Rowe and Chestnut, [\$12, \$54] in 1990\$, or [\$21, \$98] in 2015\$. In previous benefit analyses, the EPA only used the unit value for asthma exacerbation in children from Rowe and Chestnut (1986) for avoiding a “bad asthma day”. There are two other unit values for children and two unit values for adults included in BenMAP discussed further below.

The first unit value for adult is based on willingness to pay to avoid an asthma exacerbation from four WTP estimates from Rowe and Chestnut (1986) for avoiding a “bad asthma day.” The second unit value for adults was derived by using Model 1, Table III in Dickie and Ulery (2002) -- the same model used for acute bronchitis, LRS, and URS -- assuming that an asthma exacerbation consists of 1 symptom-day. As noted above, this model relates parental WTP to the number of symptom-days avoided and to whether it is the parent or the child at issue. The unit value derived from this model for adults is \$74.

Two additional unit values are available for children. One of these is twice the original unit value, or \$104, based on the evidence from Dickie and Ulery (2002) that parents are willing to pay about twice as much to avoid symptoms and illness in their children as in themselves. The third unit value is based on Model 1, Table III in Dickie and Ulery (the same model used for asthma exacerbation in adults, only now with the “adult or child” variable set to 1 rather than 0). The unit value derived from this model is \$221.

Table I-7. Additional Unit Values Available for Asthma-related Acute Symptoms and Illnesses

Health Endpoint	Basis for Estimate *	Age Range		Unit Value	Unit Value Distribution	Parameters of Distribution	
		Min	Max			P1	P2
Asthma Attacks; Cough; Moderate or Worse; One or more symptoms; Shortness of Breath; Wheeze	Bad asthma day, Rowe and Chestnut (1986)	18	99	\$59	Uniform	21.42	97.56
	1 symptom-day, Dickie and Ulery (2002)	18	99	\$104	Lognormal	4.64	0.0957
	2 x bad asthma day, Rowe and Chestnut (1986)	0	17	\$118	Uniform	42.84	195.12
	1 symptom-day, Dickie and Ulery (2002)	0	17	\$221	Lognormal	5.39	0.0925
Cough; Moderate or	Bad asthma day, Rowe and Chestnut (1986)	0	17	\$59	Uniform	21.42	97.56